

Trip Simulation and Fuel Card Modality

Service Tracking MS - Functional Documentation

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This document explains how Trip Simulation estimates revenue, route cost, fuel quantity, fuel value, trip expenses, profit, and fuel card movements. It is intended for operations, finance, procurement, and system users.

Source areas reviewed: Trip Simulation, Trip Settings, Simulation Routes, Fixed Expenses, Fuel Card, Fuel Card Recharge, Fuel Card Issue, Fuel Card Ledger Entry, Project customer details, Quotation creation, Purchase Order creation, and Fuel Card Movement Report.

1. Executive Summary

Trip Simulation is the planning document used to estimate whether a trip is commercially viable before quotation and purchasing work begins. Users enter the expected revenue manually, select the route, vehicle, project, customer, fuel card, and supplier details, and the system calculates cost and margin.

After a Trip Simulation is submitted, the system can create a Quotation and Purchase Orders. If a Fuel Card is selected, submitting the simulation deducts the trip fuel litres and monetary value from that Fuel Card through a Fuel Card Ledger Entry.

2. Core Trip Simulation Formulas

Area	Formula	Meaning / Source
Days in Trip	Return Date - Departure Date + 1	Inclusive day count. Saving is blocked if Return Date is before Departure Date.
Total Distance (km)	Sum of all Fuel table Distance rows	Route distances are loaded from Simulation Routes trip steps into the Fuel child table.
Fuel Used (Ltr)	Distance x Truck Type Litres per KM	Litres per KM comes from Trip Settings based on the selected Vehicle truck type.
Total Fuel Used (Ltr)	Sum of Fuel Used (Ltr) rows	Stored as the total trip fuel quantity requirement.
Fuel Price	Latest purchase rate for Fuel Item	Priority is submitted Purchase Invoice Item first, then submitted Purchase Order Item.
Total Fuel Cost	Total Fuel Used (Ltr) x Fuel Price	Monetary value required from the Fuel Card or fuel procurement process.
Total Trip Cost	Total Fuel Cost + Sum of Trip Expenses Outline Amount	Combines fuel and all calculated trip expense rows.
Net Profit	Expected Revenue - Total Trip Cost	Expected Revenue is manually entered by the user.
Net Profit Margin %	Net Profit / Expected Revenue x 100	Stored in net_profit_. If revenue is zero, margin becomes zero.

Important: $100 - ((\text{Expected Revenue} - \text{Total Trip Cost}) / \text{Expected Revenue}) \times 100$ is a cost ratio. The system now uses Net Profit Margin: $\text{Net Profit} / \text{Expected Revenue} \times 100$.

3. Trip Settings

Setting	Default / Type	Used For
Management Fee Percentage	3%	Management Fee = Expected Revenue x Management Fee % / 100.
Salaries Percentage	10%	Salaries = Expected Revenue x Salaries % / 100.
Driver Mileage Per Day	Currency, default 0	Driver Mileage = Driver Mileage Per Day x Days in Trip.
Heavy Truck Vehicle Cost	85,000,000	Depreciation for Heavy Truck vehicles.
Light Truck Vehicle Cost	45,000,000	Depreciation for Light Truck vehicles.
Heavy/Light Truck Tyre Price	Currency	Tyres formula by truck type.
Heavy/Light Truck Number of Tyres	Float	Tyres formula by truck type.
Heavy/Light Truck Tyre Lifecycle (km)	Float	Tyres formula by truck type.
Heavy/Light Truck Litres per KM	Float	Fuel consumption formula by truck type.

Trip Settings is a Single DocType. It stores the calculation constants centrally so users can change percentages, truck costs, tyre assumptions, fuel ratios, and driver mileage without editing code.

4. Expense Formulas

Expense	Formula	Notes
Management Fee	Expected Revenue x Management Fee % / 100	Percentage is read from Trip Settings. Quantity stores the percentage; rate is Expected Revenue / 100.
Salaries	Expected Revenue x Salaries % / 100	Percentage is read from Trip Settings. Default is 10%.
Driver Mileage	Driver Mileage Per Day x Days in Trip	Daily amount is read from Trip Settings, not Fixed Expenses.
Maintenance Fee	Previous Month Maintenance Cost / 30 x Days in Trip	Previous month maintenance cost is read from submitted Purchase Invoice Items linked to Purchase Orders and EAH Job Cards for the selected vehicle.
Tyres	Tyre Price x Number of Tyres / Tyre Lifecycle KM x Total Distance KM	Tyre settings are read from Trip Settings by vehicle truck type. Tyres are exempt from route predefined amount restriction.
Depreciation	Vehicle Cost / Month Number / 12 / 30 x Days in Trip	Vehicle Cost comes from Trip Settings by truck type. Month Number is the month number of the departure/transaction date.
Per Trip Day	Route Amount x Days in Trip	Used for ordinary Fixed Expenses configured as Per Trip Day.
Salary Allocation	Salaries Constant / 30 / Active Vehicles x Days in Trip	Legacy calculation method remains supported where configured.
Percentage of Expected Revenue	Expected Revenue x Expense % / 100	Uses the percentage configured on Fixed Expenses or the row quantity.
Fixed Amount	Route Amount	Used when no special calculation method applies.

5. Maintenance Cost Modality

Maintenance Fee uses actual maintenance history instead of a manually typed monthly constant. The system looks at the month before the trip reference date and sums submitted Purchase Invoice Item values that are linked to Purchase Orders. Those Purchase Orders must be linked to EAH Job Cards, and the EAH Job Card vehicle must match the Trip Simulation vehicle.

Maintenance Fee = Previous Month Maintenance Cost / 30 x Days in Trip

The Trip Expenses child table stores the previous month maintenance cost in a read-only field so users can see the source cost used by the formula and reports can expose it.

6. Fuel Calculation Modality

Step	Calculation / Source	Result
1	Select Vehicle	System identifies Vehicle truck type, for example Heavy Truck or Light Truck.
2	Read Trip Settings	System gets the matching litres per KM field for that truck type.
3	Load Simulation Route	Each route step distance is copied into the Fuel child table.
4	Calculate row litres	Fuel Used (Ltr) = Distance x Litres per KM.
5	Calculate total litres	Total Fuel Used (Ltr) = Sum of row fuel litres.
6	Fetch fuel price	Last submitted Purchase Invoice Item rate for Fuel Item, else last submitted Purchase Order Item rate.
7	Calculate total fuel value	Total Fuel Cost = Total Fuel Used (Ltr) x Fuel Price.

Example: if a Heavy Truck uses 0.300 litres per KM and the route distance is 572 KM, Total Fuel Used = 572 x 0.300 = 171.6 litres.

7. Fuel Card Modality

The Fuel Card feature models prepaid fuel. Procurement can recharge a card with litres and value, while trips and other fuel issues consume litres and value. The Fuel Card Ledger Entry is the source of truth for all balance movements.

Transaction	Movement	Formula / Control
Fuel Card Opening Balance	Initial litres and value	Stored on Fuel Card as opening_litres and opening_value.
Recharge	Litres In	Amount = Litres x Rate. On submit, creates a Recharge ledger entry.
Trip Usage	Litres Out	On Trip Simulation submit, Total Fuel Used is deducted at Fuel Price.
Other Issue	Litres Out	Fuel Card Issue handles office car or non-simulation fuel usage. Amount = Litres x Rate.
Cancellation of Recharge	Litres Out	Cancelling a recharge creates a reversing Cancellation entry.
Cancellation of Issue / Trip	Litres In	Cancelling an issue or simulation creates a reversing Cancellation entry.

Ledger Amount = Litres In or Litres Out x Rate

Balance Litres After Transaction = Previous Balance + Litres In - Litres Out

Balance Value After Transaction = Previous Value + Value In - Value Out

The system blocks a ledger entry if it would make the card litres negative or the monetary value negative. For Trip Simulation, this means a trip cannot consume more fuel value than the selected Fuel Card has available. Users are asked to refill the Fuel Card.

8. Fuel Card Dashboard on Trip Simulation

When a Fuel Card is selected in Trip Simulation, the form shows a fuel card status area. It compares the requested trip fuel amount against the current Fuel Card balance. This helps the user know whether the trip can be submitted or whether the card should be recharged first.

Dashboard Value	Meaning
Card Balance Litres	Current available litres from opening balance plus ledger movements.
Card Balance Value	Current available monetary value from opening value plus ledger movements.
Requested Litres	Trip Simulation Total Fuel Used (Ltr).
Requested Amount	Total Fuel Used (Ltr) x Fuel Price.
Availability Status	Shows whether the selected card has enough monetary value for the trip.

9. Document Creation and Linkage

Document	When Created	Linkage
Quotation	After Trip Simulation is submitted using Create Quotation	Quotation stores custom_trip_simulation where the custom field exists; Trip Simulation stores the Quotation reference.
Fuel Purchase Order	After Trip Simulation is submitted using Create Purchase Order for fuel	Purchase Order stores custom_trip_simulation and custom_vehicle where fields exist; Trip Simulation stores fuel_purchase_order.
Expense Purchase Order	For selected payable expenses	Only Fixed Expenses marked Is Payable can be converted to Purchase Order items.
Fuel Card Ledger Entry	Automatically on submit/cancel	Trip Simulation stores fuel_card_ledger_entry for the submit deduction.

10. Project Customer Details

The Project form contains Customer Details used by the Trip Simulation and quotation flow. These include Customer, Route, Sales Order, Rate per trip, Sales Item, Driver Mileage per Trip, Fuel Entitlement Per Trip, Fuel Item, and Mileage Item.

Sales Item is especially important for Create Quotation. Trip Simulation uses Expected Revenue as the Quotation rate, but the item code is read from Project Sales Item.

11. Controls and Validations

Control	Behavior
Targeted Net Profit	Saving/submitting is blocked when Net Profit Margin is below Targeted Net Profit.
Duplicate Expenses	Duplicate expense rows are blocked after normalizing labels with trim/lowercase and standard labels.
Route Expense Limits	Expenses cannot exceed their route predefined amount, except Tyres.
Custom Expenses	Route expense labels are stored as data so operational custom labels can be added, but payable purchase orders still require matching Fixed Expenses with an Item.
Fuel Price	Trip Fuel Card deduction requires Fuel Price greater than zero.
Fuel Card Balance	Ledger entries are blocked if litres or monetary balance would become negative.
Ledger Integrity	Fuel Card Ledger Entry cannot be deleted; users must create reversing entries.
Submitted Document Cleanliness	Post-submit link updates use db_set/db_set_value with update_modified=False where needed to avoid unnecessary Not Saved state.

12. Reports and Workspace

Area	Shows
Trip Simulation Center Workspace	Number cards for simulations, revenue, cost, profit/loss, fuel card balance litres, and fuel card balance value.
Workspace Charts	Revenue vs Cost vs Profit/Loss chart and Trip Cost Breakdown pie chart.
Trip Analysis Report	Columns include revenue, total trip cost, fuel cost, fuel litres, expense breakdown, net profit, net margin, and margin variance.
Fuel Card Movement Report	Recharge, trip usage, other issues, rate, litres in/out, amount, and running balances.

13. Quick Formula Reference

Days in Trip = Return Date - Departure Date + 1

Fuel Used (Ltr) = Distance KM x Truck Type Litres per KM

Total Fuel Cost = Total Fuel Used (Ltr) x Fuel Price

Management Fee = Expected Revenue x Management Fee % / 100

Salaries = Expected Revenue x Salaries % / 100

Driver Mileage = Driver Mileage Per Day x Days in Trip

Maintenance Fee = Previous Month Maintenance Cost / 30 x Days in Trip

Tyres = Tyre Price x Number of Tyres / Tyre Lifecycle KM x Total Distance KM

Depreciation = Vehicle Cost / Month Number / 12 / 30 x Days in Trip

Total Trip Cost = Total Fuel Cost + Sum of Trip Expense Amounts

Net Profit = Expected Revenue - Total Trip Cost

Net Profit Margin % = Net Profit / Expected Revenue x 100

Fuel Card Amount = Litres x Rate

Fuel Card Litres Balance = Opening Litres + Sum(Litres In - Litres Out)

Fuel Card Value Balance = Opening Value + Sum((Litres In - Litres Out) x Rate)